

Ali Abhari
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Highlights of Qualifications

- Strong **mechatronics, control system** and **simulation** environment development skills.
- Extensive embedded controller software development and testing experience through model based development, simulation, **MIL, SIL, HIL** and in vehicle validation, development for Software defined Vehicle (**SDV**) and Brake by Wire (**BbW**).
- Highly skilled in technical computer software **MATLAB, SIMULINK, CarSim, ASCET, INCA, Doors, CANalyzer, CANape, Integrity, Quality Center (ALM), VB.net** and C.
- Competent in different areas of Engineering and Computer Science; Mechanical Design, Solid Mechanics, Dynamics, Vibrations, Controls, Finite Elements Analysis, Machine Learning, (Deep) Neural Networks, Image Processing, NLP, ...

Professional Experience

Subsystem Lead / Modeling Engineer

GM

2021-2024

- Developed system health monitoring requirements for brake-by-wire (**BbW**) electronic brake subsystems (**SDV** architecture).
- Updated anode subsystem models of the Fuel Cell to the new generation under development (**SIMULINK**).
- Developed test scenarios in Simian for validating the automated driving features requirements.

Principal Software Engineer - Functions & Controls

ZF-TRW

2018-2020

- Developed and adapted software functions for automotive **Electric Power Steering (EPS)** (**SIMULINK, TargetLink**).
- Tested and debugged the developed algorithms in the unit testing, SIL, HIL and in vehicle environments.
- Performed stability analysis for the developed control software through System Identification, Correlation Analysis and stability on the released versions of the software for in-vehicle tests.

Sr. Software Engineer - Simulations

Bosch

2011-2018

- Designed, Modified and Integrated control software for automotive **Electronic Stability Control (ESC)** using **ASCET**.
- Developed simulation models for vehicle system representation and tests using Simulink, employing internal and external vehicle dynamics models (CarSim)
- Employed all components of **V-model** development including requirement engineering, design and review documentation.
- Tested developed control algorithm for **Vehicle Dynamics Controller** through compiling, flashing and performing highly dynamic maneuvers in simulation, HIL benches, test tracks and winter tests.
- Developed and performed fully automated simulated integration tests to verify compliance with the requirements.

Controls Integration Systems Engineer

Honeywell

2009-2010

- Defined stability test plans, disturbance sources and induced effects for **Airbus A350** Environmental Control System (**ECS**).
- Identified Cross-Work-Package **stability** requirements, implemented through **Doors** requirement management system.

Simulation and Controls Engineer

DaimlerChrysler

2007-2009

- Changed and debugged existing control models for HEV transmission controller using Matlab/Simulink and RTW.
- Performed **hardware-in-the-loop (HIL) real-time** simulations to calibrate models parameters using **INCA**.
- Performed in-vehicle tests for developed control models, collected and analyzed signals traces from the **CAN** bus.
- Proposed to and Implemented recommendations by the fuel economy taskforce, tested for their effectiveness.

Device Development Engineer

Sunnybrook

2005-2007

- Designed a new small scale robotic arm for MRI imaging navigation and guidance using **Solid Works**.
- Performed analytical **forward and inverse dynamics modeling** of the robotic arm, Implemented the model in **Simulink**.

Education

Ph.D. Mechanical Engineering (Mechatronics) - [University of Toronto]

M.Sc. Mechanical Engineering (Applied Mechanics & FEA) - [University of Tehran]

BSME Mechanical Engineering (Solid Mechanics & Design) - [Sharif University of Technology]